



We are in the process of migrating all NASA Earth science data sites into Earthdata from now until end of 2026. Not all NASA Earth science data and resources will appear here until then. Thank you for your patience as we make this transition. [Read about the Web Unification Project](#)



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EARTHDATA

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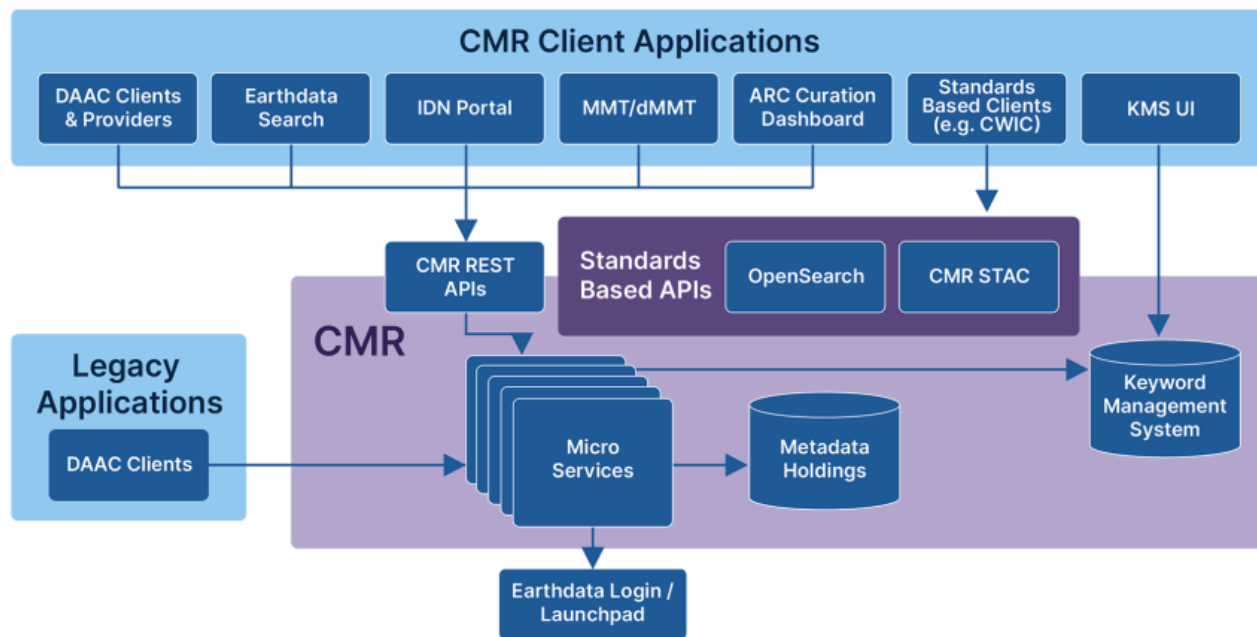
Common Metadata Repository (CMR)

CMR is the authoritative management system for all Earth Observing System Data and Information System (EOSDIS) metadata.

NASA's Common Metadata Repository (CMR) is a high-performance, high-quality, continuously evolving metadata system that catalogs all data and service metadata records for NASA's Earth Observing System Data and Information System ([EOSDIS](#)) and will be the authoritative management system for all EOSDIS metadata. These metadata records are registered, modified, discovered, and accessed through programmatic interfaces leveraging standard protocols and APIs.

CMR is the keystone that makes NASA's Earth observation data discoverable. As a metadata repository, CMR contains Unified Metadata Model (UMM) schema records that describe individual Earth data files (UMM-Granules), collections of files (UMM-Collections), scientific details about the data files (UMM-Variables), related tools and services that act on the data files (UMM-Tools and -Services), and pertinent relationships between these concepts. Using the UMM allows CMR to host its metadata records in several supported native formats, with translation services available between formats.

As of 2024, CMR contains metadata describing over a billion Earth Observation data files stemming from approximately 10,000 data collections, but that just comprises NASA's Earth Science Holdings. Because NASA is a member of the Committee on Earth Observation Satellites ([CEOS](#)), CMR also houses collection records for the dozens of other CEOS member agencies which contribute another 45,000 collection metadata records to CMR. Additionally, if our CEOS partners provide OpenSearch Description Documents (OSDDs) for their collection metadata records, the data files they house within those collections are also made discoverable through CMR, enabling interagency and international federated discovery. OpenSearch is an Open Geospatial Consortium (OGC) international standard for federated search and discovery.



CMR also has application programming interfaces (APIs) for both creating new metadata holdings (CMR Ingest) and querying the repository's contents (CMR Search). Distributed Active Archive Centers (DAACs) primarily utilize CMR Ingest, curating and maintaining the metadata pertaining to collections that fall under their scientific purview. Anyone — from private industries to scientists to students to policymakers to anyone across the globe — can and do search CMR extensively. CMR nominally sees an average search load of several thousand queries per minute, with typical peaks over 10,000 queries per minute.

CMR tools and services include:

- [CMR Application](#)
- [Metadata Management Tool \(MMT\)](#)
- [draft Metadata Management Tool \(dMMT\)](#)
- [Metadata Curation Dashboard](#)

CMR is the backend behind the following data search and discovery services:

- [Earthdata Search](#)
- [International Data Network \(IDN\)](#)

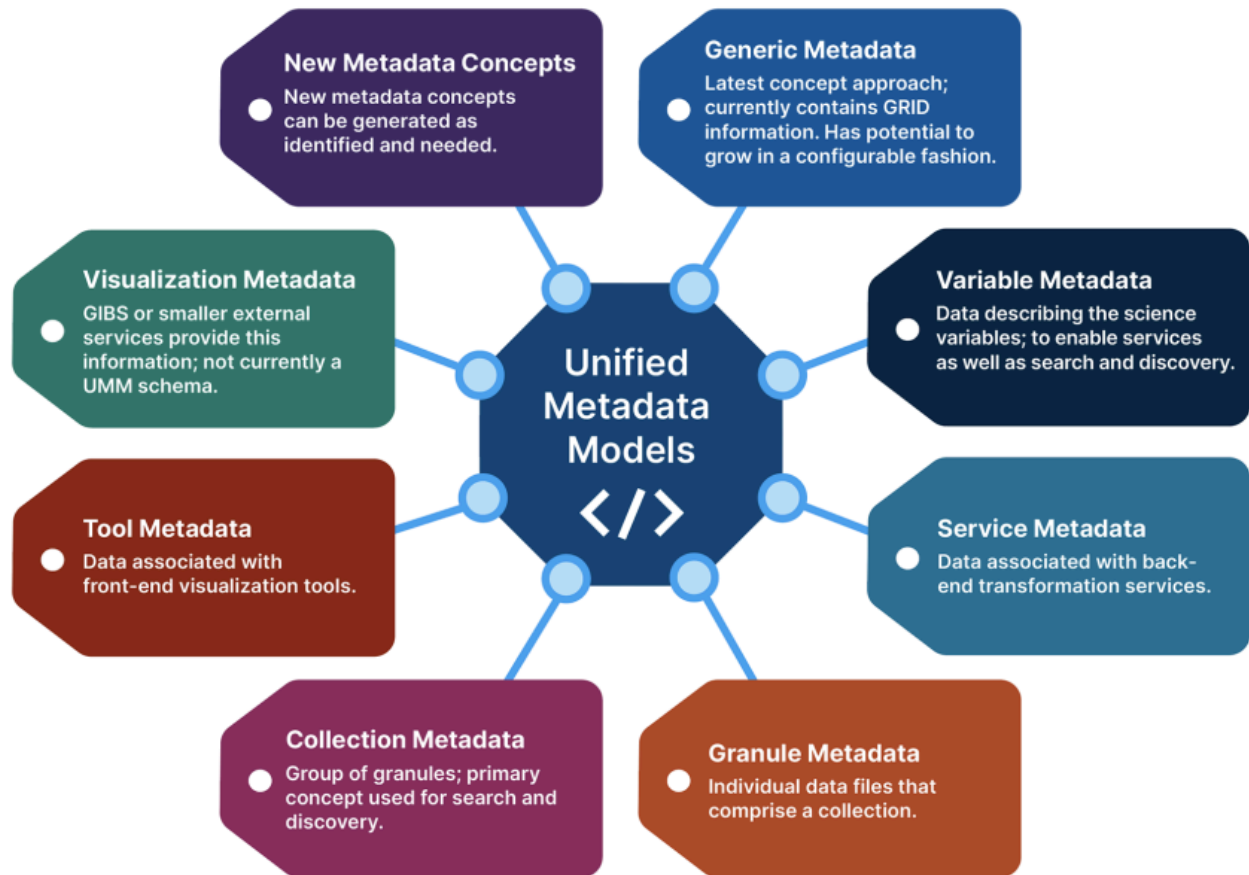
All our metadata applications and records (metadata contents) heavily rely upon the GCMD Keywords to maintain a controlled vocabulary, which is instrumental to being able to effectively search through a multitude of metadata records. These are accessible through a [Keyword Management System \(KMS\) API](#) or through a GUI [GCMD Keyword Viewer application](#), with changes being community-driven via the [Earthdata Forum](#).

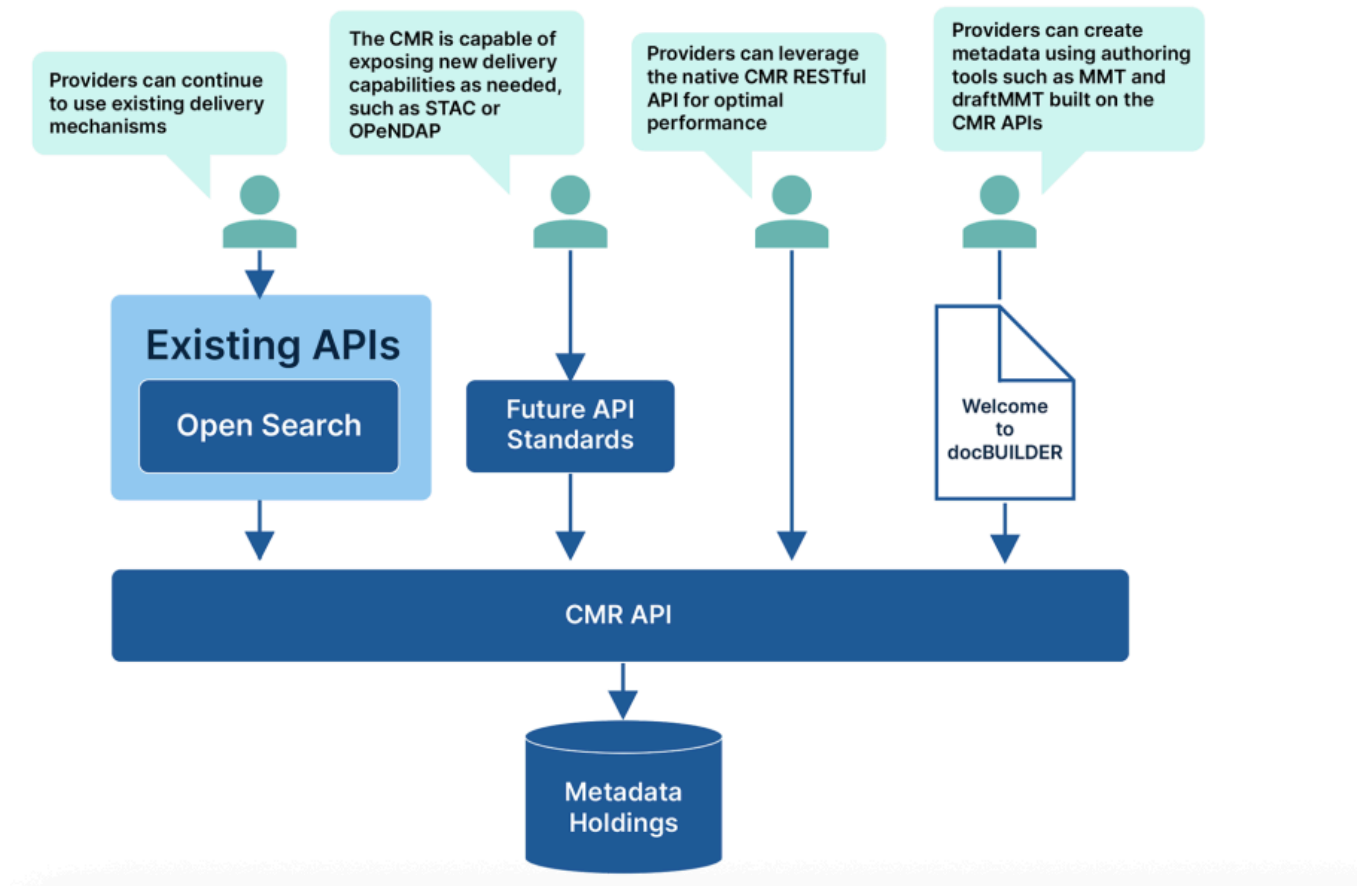
CMR Search can be performed through the native CMR API or through [CMR STAC](#), [CMR Cloud STAC](#), and [CMR OpenSearch API wrappers](#). CMR Graph Query Language (GraphQL) can also be used to find concepts and relationships. Metadata concept associations can be made directly through linkages within metadata fields, or through CMR Graph Database (GraphDB) relationships.

Goals of CMR

The primary goal of CMR is to be a highly performant tool for both search and ingest: to deliver search results (for both collection and granule records) in under a second and be able to handle the ingest, from receipt to visibility, in under an hour (though CMR does this in well under a minute) without dropping a single file or

change. Search and ingest loads have and will continue to increase over time and with the addition of new missions. Current loads typically range within thousands of concurrent search and ingest requests occurring per minute every day (with surges well into the upper tens of thousands per minute for each, when a provider reprocesses and re-ingests a large collection or when a user makes a massive request). Having an easily scalable architecture is key to remaining both performant and cost effective.





How does CMR do that?



Why CMR?



Benefits of CMR



Getting Started



Additional Information

- [Data Use and Citation Policies](#): Visit this page to get details with examples on how to on how to use, cite or reference NASA's discipline-specific data and services.
- EOSDIS Tools and Services
 - [EOSDIS Tools and Service Portfolio](#) was created to:
 - Evaluate and update the existing inventory DAAC and EOSDIS tools and services to understand the current portfolio
 - Identify tools and services with apparent overlapping functionality and evaluate to determine what, if any, efficiencies may be gained moving forward
 - Create a framework for a governance process to apply to future tool and service development efforts
 - [EOSDIS Tools Information](#) NASA's Distributed Active Archive Centers (DAACs) provide an extensive variety of [tools](#) that make it easy for users to discover, access, manipulate and use EOSDIS data

products.

Find Data



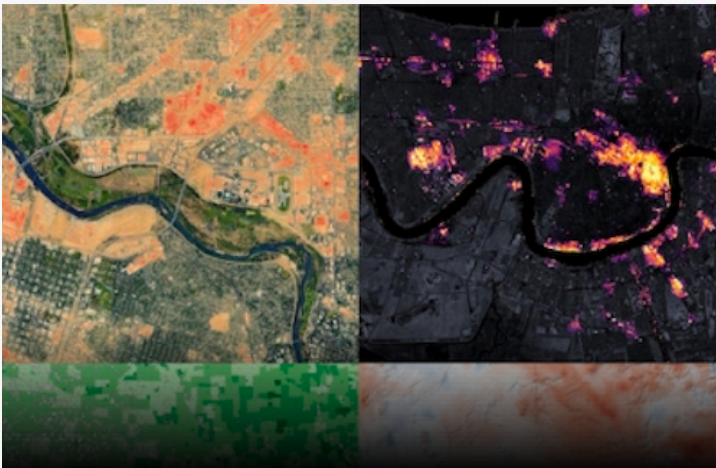
By Platform →

Explore data collected by platforms in space, air, water, and land.



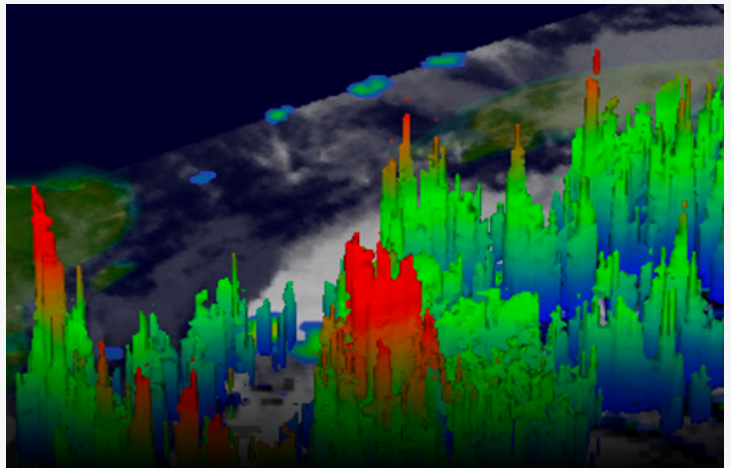
By Topic →

Browse by topic to find relevant data, tools, learning resources, news, and more.



Data Catalog →

Dive into one of the planet's largest repositories of Earth science data.



Data Tools →

Access a range of tools that enable data discovery, analysis, and visualization.



Find Data

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